MangeVision: Differentiating Sarcoptic vs Demodectic Mange

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Tier 1-My project is Tier 1 it is a simpler project built from pre-trained models.

The Problem

- The real-world problem that I am addressing is the manual diagnosis of mange. As I previously worked in a veterinary clinic, I know firsthand that the process of scraping the animal's skin is time consuming, and uncomfortable for the animal. If the sample acquired didn’t capture the mites it would require another skin scraping and more discomfort to the animal.
- Who cares about this problem would be veterinarians, veterinary staff members, and pet owners.
- This is important because sarcoptic mange is highly contagious to other animals and humans, so correct diagnosis is critical.

My Solution

My system is a vison based diagnostic tool that analyzes photos of skin lesions and detect characteristics between Sarcoptic (contagious) and Demodectic (noncontagious) mange.

- Input: High resolution clinical imagery.
- Logic: YOLOv8 Deep Learning Agents
- Output: Precision bounding boxes with diagnostic confidence scores

Technical Approach

- CV Technique: Object Detection & Classification
- Model: YOLOv8
- Framework: PyTorch
- Why: My technique uses object detection to locate specific lesions on the skin, and classification to determine the type of mange. I chose YOLOv8 because it is famous for being 'real-time,' meaning it can identify skin issues almost instantly, and I selected PyTorch because it is a beginner-friendly framework that makes it easy to experiment with different neural network settings.

Technical Approach cont.

Feature Extraction

The Observer

Identifying Raw Patterns

The system scans raw pixels to identify the unique textures of mange, such as inflammation, hair loos, and crusting.

Powered by the CSPDarknet engine for high-speed scanning.

Multi Scale Analysis

The Intelligence Layer

Contextual Awareness

This layer combines broad visual context with tiny details. It ensures the model can see a single mite lesion and large infected patch at the same time

Used FPN & PAN logic to fuse data across different zoom levels.

Diagnostic Output

The Classifier

Prediction &Localization

The final engine draws the bounding box and assigns a probability score.

Uses a Decoupled Head to separate what it is from where it is.

Dataset & Processing

- Sources:
  - https://www.kaggle.com/datasets/youssefmohmmed/dogs-skin-diseases-image-dataset
  - https://www.kaggle.com/datasets/devdope/skin-disease-raw-dataset
- Size: approximately 2,000 images for the three classes (Sarcoptic, Demodicosis, Healthy/Other)
- Labels: I will used Roboflow to annotate bounding boxes around active lesion sites.

Category	Version 1	Version 2 (Final)
Unique Images	142	209
Data Intergrity	Initial Baseline (Mixed Duplicates)	Focused Duplicate Filtering
Augmentation	Flip/90 Rotate	Flip/90 Rotate
Validation Split	Verified 10% Split	Verified 10% Split

Live Demo

Here is my YouTube link to my demo video:

https://youtu.be/PpDsgj9_If4

Results

- Demodicosis Accuracy(mAP50): 92%
- Scabies Accuracy(mAP50): 38%
- Overall Accuracy(mAP50): 65%
- Inference Speed: 3.2 ms

```
25 epochs completed in 0.029 hours.
Optimizer stripped from /content/runs/detect/train/weights/last.pt, 6.2MB
Optimizer stripped from /content/runs/detect/train/weights/best.pt, 6.2MB

Validating /content/runs/detect/train/weights/best.pt...
Ultralytics 8.4.46 🚀 Python-3.12.13 torch-2.10.0+cu128 CUDA:0 (Tesla T4, 14913MiB)
Model summary (fused): 73 layers, 3,006,038 parameters, 0 gradients, 8.1 GFLOPs
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 6.1it/s 0.2s
                   all         20         20      0.488      0.767       0.65      0.366
           Demodicosis          5          5      0.529          1       0.92      0.574
               Scabies         15         15      0.447      0.533      0.379      0.157
Speed: 0.2ms preprocess, 1.9ms inference, 0.0ms loss, 1.4ms postprocess per image
Results saved to /content/runs/detect/train
```

Results-Visual Validation

- Success-blue bounding boxes are tight around the lesions

- Failure-the model failed detecting this image as the lesions differ than what the majority of the other lesions the Scabies images.

Key Learning

- High mAP scores can be misleading is a specific class in my case Scabies is under performing. Analyzing individual “No Detection” logs was critical for identifying class bias.
- Demodicosis lesions are visually louder. Future iterations require targeted over sampling of subtle rash style images to balance sensitivity.

Future Work

- Dataset Expansion- Obtain images of Scabies lesions from canines
- Multi-Species Diagnostics- Expand training to include mange variations in livestock and wildlife.
- Clinical Integration- Develop a Confidence Threshold alert system that suggests a manual skin scraping if the AI’s confidence falls below 75%.

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